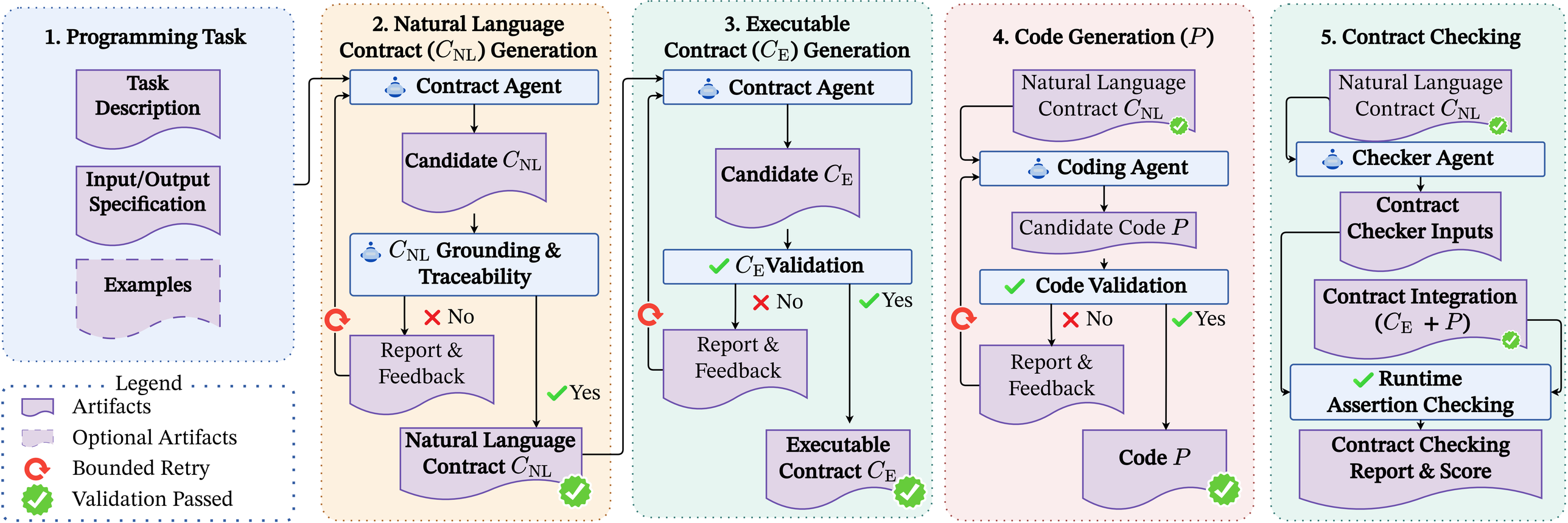

1. Programming Task
Task Description
Input/Output Specification
Examples
Legend
Artifacts
Optional Artifacts
Bounded Retry
Validation Passed
2. Natural Language Contract (C_{NL}) Generation
Contract Agent
Candidate C_{NL}
C_{NL} Grounding & Traceability
No
Report & Feedback
Yes
Natural Language Contract C_{NL}
3. Executable Contract (C_E) Generation
Contract Agent
Candidate C_E
C_E Validation
No
Report & Feedback
Yes
Executable Contract C_E
4. Code Generation (P)
Natural Language Contract C_{NL}
Coding Agent
Candidate Code P
Code Validation
No
Report & Feedback
Yes
Code P
5. Contract Checking
Natural Language Contract C_{NL}
Checker Agent
Contract Checker Inputs
Contract Integration (C_E + P)
Runtime Assertion Checking
Contract Checking Report & Score